

EXAM POV NOTES (Q&A FORMAT)

10-MARK QUESTIONS

Q1: Compare and contrast the three types of wireless network topologies (Point-to-Point, Multipoint-to-Point, and Point-to-Multipoint). Include descriptions of their applications, associated security challenges, and examples of their common usage (such as WLANs and cellular networks).

Wireless network topologies define the fundamental arrangements by which devices communicate using media composed of electromagnetic (EM) waves or infrared waves.

- **2. Point-to-Point (p2p) Definition:** This topology is the preferred method for communication when only two wireless devices need to exchange data directly.
- **3. p2p Applications & Usage:** Common applications include using infrared (IR) waves for remote controls communicating with controlled devices, or using radio waves (like Bluetooth) in Internet-of-Things (IoT) scenarios where direct communication is needed.
- **4. p2p Operational Limitations:** P2P networks are typically characterised as being slow because they operate in **half duplex mode**, meaning only one device can transmit data at any given time.
- **5. p2p Security Concern:** A key security disadvantage is the risk that a source device could push malicious content into the target device without the target device being able to detect it.
- **6. Multipoint-to-Point (mp-2-p) Definition:** This is the most prevalent topology used for **Wireless Local Area Networks (WLANs)**, where multiple user devices contend to communicate with a Wireless Access Point (WAP) over a shared frequency band.
- **7. mp-2-p Usage & Standard Name:** The wireless LAN standard term for this arrangement is the **Basic Service Set (BSS)**, commonly used for setting up networks in small areas (less than 2,500 square feet), such as houses or small buildings.
- **8. mp-2-p Operational Feature:** The mp-2-p topology employs **Frequency Division Multiplexing (FDM)**, which helps manage bandwidth efficiently through the use of non-overlapping frequency bands.
- **9. Point-to-Multipoint (p2mp) Definition & Usage:** This topology involves strategically placing multiple WAPs to cover a significant area, a setup known as an **Extended Service Set (ESS)**, and is used by all cellular networks.
- **10. p2mp Security and Cost:** These networks are costly to set up and maintain. Security is a significant concern, particularly when these networks utilise the IEEE 802.11 standard.

Feature	Point-to-Point (p2p)	Multipoint-to-Point (mp-2-p)	Point-to-Multipoint (p2mp)
Standard Use	IR Remotes, Bluetooth (IoT)	WLAN (BSS), Small Buildings	Cellular Networks, ESS
Operational Mode	Half Duplex (Slow)	Uses FDM for Bandwidth	Multiple WAPs (Coverage)
Security	Malicious Injection Risk	Schemes are generally "hack-able"	Security is a major concern

Q2: Describe the evolution of wireless security protocols defined by the Wireless Alliance (WEP, WPA, WPA2, and WPA3). Detail the specific weaknesses WPA was designed to address in WEP, and elaborate on three noteworthy features that WPA3 introduced to enhance security over WPA2.

The Wireless Alliance has defined a series of authentication security protocols—WEP, WPA, WPA2, and WPA3—to secure 802.11 wireless networks.

- **2. WEP (Wired Equivalent Privacy):** WEP was the initial protocol, designed in 1997, which used a combination of system- and user-generated key values for data encryption.

- 3. **WEP Weakness:** WEP proved to be the least secure option because hackers developed strategies to reverse-engineer and break the protocol's encryption mechanism.
- 4. **WPA Introduction & Purpose:** Wi-Fi Protected Access (WPA) was created specifically as a response to address the known weaknesses inherent in the older WEP protocol.
- 5. **WPA Improvement 1 (Dynamic Key):** WPA introduced the **Temporal Key Integrity Protocol (TKIP)**, which employs a dynamic, 128-bit key that proved substantially more difficult to crack than WEP's static, unchanging key.
- 6. **WPA Improvement 2 (Integrity Check):** WPA also featured a **Message Integrity Check (MIC)**, which served to look for and identify any tampered data packets transmitted by hackers.
- 7. **WPA3 Feature 1 (Stronger Brute Force Defense):** WPA3 significantly enhances security by defending against **offline password guesses**. It limits users to one guess per attempt and forces them to engage directly with the Wi-Fi equipment, requiring physical presence for each guess.
- 8. **WPA3 Feature 2 (SAE Protocol):** WPA3 incorporates the **Simultaneous Authentication of Equals (SAE) protocol** to provide a secure handshake between a device and the AP, ensuring verified authentication and connection even if the user has a weak password.
- 9. **WPA3 Feature 3 (Individualized Data Encryption/DPP):** For public networks, WPA3 replaces shared passwords with mechanisms like the **Wi-Fi Device Provisioning Protocol (DPP)**, which allows devices to be added to the network securely using NFC tags or QR codes.
- 10. **WPA3 Feature 4 (Encryption Level):** WPA3 implements **GCMP-256 encryption**, providing a higher level of cryptographic security than the 128-bit encryption standard previously used.

Q3: Explain the architecture and operational framework of IEEE 802.11 networks. Differentiate between the two main operating modes (Infrastructure BSS and Independent BSS) and explain the roles of the Access Point (AP) and the Distribution System (DS).

The IEEE 802.11 Committee defined protocols for Wireless Local Area Networks (WLANs), providing features for wireless media, similar to how Ethernet provides services for wired media.

- 2. **MAC Layer Support:** The 802.11 architecture's basic services for WLANs are primarily implemented and supported by the MAC layer.
- 3. **Access Point (AP) Role:** The AP acts as a bridge, supporting both wired and wireless connections. It bridges the wireless and wired Layer 2 (L2) Ethernet frames.
- 4. **Distribution System (DS) Role:** The 802.11 standard refers to the upstream wired network that the AP connects to as the **Distribution System (DS)**. The AP allows traffic to flow between the wireless network and the DS.
- 5. **BSS Categorization:** The Basic Service Set (BSS) is the fundamental unit of operation and is categorised into different types based on the mode of operation.
- 6. **Infrastructure BSS Definition:** In this mode, wireless stations must communicate through the Access Point.
- 7. **Infrastructure BSS Formation:** The coverage area of an Infrastructure BSS is collectively defined by the AP and all its associated wireless clients.
- 8. **Independent BSS Definition:** This mode supports direct, mutual communication between the individual wireless clients themselves, without relying on an AP.
- 9. **Independent BSS Characteristics:** An Independent BSS is spontaneously created and is commonly referred to as an **ad-hoc network**.
- 10. **Independent BSS Limitation:** A critical limitation of Independent BSS is that it **does not support access** to wired networks.

Q4: Detail the primary functions of the four core layers of the Bluetooth Protocol Stack: Radio (RF) Layer, Baseband Link Layer, Link Manager Protocol Layer, and Logical Link Control and Adaption (L2CAP) Protocol Layer.

The Bluetooth Protocol Stack governs how devices use radio waves to transmit and receive data, managing connections within the basic architecture unit known as a piconet.

- **2. Radio (RF) Layer Function 1 (Air Interface):** This layer defines the air interface details, including frequency, the physical characteristics of Bluetooth transceivers, and the use of frequency hopping and transmit power.
- **3. Radio (RF) Layer Function 2 (Modulation):** Its primary task is to perform the modulation and demodulation of data, converting digital data into RF signals and vice versa.
- **4. Baseband Link Layer Function 1 (Digital Engine/MAC):** The Baseband is considered the digital engine of the Bluetooth system and is functionally equivalent to the MAC sublayer found in LANs.
- **5. Baseband Link Layer Function 2 (Connection Establishment):** It executes crucial connection setup tasks within a piconet, handling addressing, timing, power control, and defining the packet format.
- **6. Link Manager Protocol (LMP) Function 1 (Link Management):** The LMP is responsible for the ongoing management and monitoring of links that have already been established.
- **7. Link Manager Protocol (LMP) Function 2 (Security):** This layer performs authentication and encryption processes, and is also tasked with creating links, monitoring their health, and terminating them gracefully.
- **8. Logical Link Control and Adaption (L2CAP) Definition:** L2CAP is referred to as the **heart of the Bluetooth protocol stack** because it enables communication and interaction between the upper and lower layers.
- **9. L2CAP Function 1 (Data Packaging):** It packages data packets received from the upper layers into the appropriate form expected by the lower layers of the stack.
- **10. L2CAP Function 2 (Adaption Services):** The L2CAP layer performs essential adaptation services, including both **segmentation** (breaking down large packets) and **multiplexing** (combining data streams).

5-MARK QUESTIONS

Q5: Define an Antenna and explain its function as a transducer. Describe the difference between a transmitting antenna and a receiving antenna, and note when the same antenna can be used for both roles.

An Antenna (or Aerial) is defined as a **transducer**.

- **2. Transducer Function:** Its fundamental function as a transducer is to convert electrical power into electromagnetic waves, and conversely, convert electromagnetic waves back into electrical power.
- **3. Transmitting Antenna:** A transmitting antenna operates by converting electrical signals into electromagnetic waves, which it then radiates away from the device.
- **4. Receiving Antenna:** A receiving antenna performs the opposite action, converting the electromagnetic waves collected from the incoming signal beam into usable electrical signals.
- **5. Dual Role Capability:** In systems that facilitate two-way communication, the **same physical antenna** can be efficiently used for both the transmission and the reception of signals.

Q6: Explain the concept of an Access Control List (ACL). Differentiate between Filesystem ACLs and Networking ACLs, describing the role each plays in managing access.

An Access Control List (ACL) is a set of rules used in network security to either permit or deny access to a specific computer environment, functioning much like a secure guest list.

- 2. **Overall Role:** ACLs enable administrators to ensure that a device cannot gain entry unless it presents the proper credentials.
- 3. **Filesystem ACLs Definition:** These ACLs operate as internal filters, specifically managing access to files or directories within a system.
- 4. **Filesystem ACLs Role:** They instruct the operating system on which users are permitted to access the system and define the precise privileges (e.g., read, write) those users are granted once they are inside.
- 5. **Networking ACLs Role:** These ACLs manage access to the network itself by instructing routers and switches about the types of traffic allowed to interface with the network, as well as dictating what each user or device can do once connected.

Q7: What is the Real Time Transport Protocol (RTP)? Describe its operational constraints (e.g., requirement to be used with UDP) and list four common applications of RTP.

The Real Time Transport Protocol (RTP) is a specialised protocol designed for handling real-time traffic across the Internet, such as audio and video streams.

- 2. **UDP Constraint:** A key operational constraint is that RTP must be utilised in conjunction with the **User Datagram Protocol (UDP)**.
- 3. **Delivery Mechanism Lack:** RTP does not possess its own built-in delivery mechanisms, such as defined port numbers or support for multicasting.
- 4. **Application 1: VoIP:** RTP is widely used for Voice over Internet Protocol (VoIP).
- 5. **Applications 2, 3, 4:** Other common applications include Video Teleconferencing over the Internet, Internet audio and video streaming, and assisting with tasks like media mixing, sequencing, and time-stamping.

Q8: Outline the key considerations that must be evaluated when performing the physical design of a wireless network, including environment evaluation and capacity planning.

Physical design involves planning and implementing the necessary infrastructure to meet a business's operational needs.

- 2. **Site Survey:** A crucial step is to conduct an RF (Radio Frequency) site survey to assess the environment's characteristics, identifying potential RF interference and determining accurate coverage areas.
- 3. **Physical Environment Evaluation:** The physical environment must be evaluated, including an assessment of building materials, to help eliminate sources of interference.
- 4. **Device Placement and Cabling:** Planning includes determining the type of cabling infrastructure required (wired or wireless) and strategically placing network devices like routers and Access Points (APs) to ensure seamless network coverage.
- 5. **Capacity Planning:** This step involves careful consideration of the network's potential load, including the total number of active devices, the types of devices, typical usage patterns, the applications that will be used, and the overall area that requires coverage.

Q9: Explain the roles of the Sequence Number and the Time Stamp fields within the RTP Header Format, noting how they help in sequencing and defining temporal relationships between packets.

The RTP Header Format contains specific fields essential for managing the flow and timing of real-time packets.

- **2. Sequence Number Definition:** This is a 16-bit field used to assign unique serial numbers to each RTP packet.
- **3. Sequence Number Function:** Its function is to help in sequencing the packets correctly. The number for the first packet is chosen randomly, and then the sequence number for every subsequent packet is incremented by 1.
- **4. Sequencing Help:** This field is primarily useful for checking for lost packets and detecting order mismatch (when packets arrive in the wrong sequence).
- **5. Time Stamp Function:** The Time Stamp is a 32-bit field used to define the relationship between the times of different RTP packets. Its value is calculated by adding the previous timestamp to the time taken to produce the first byte of the current packet.

Q10: Discuss the management and allocation of radio frequencies worldwide. Explain the role of the ITU-R and differentiate between licensed and license-free frequencies, providing specific examples of license-free bands.

Radio frequencies are utilised globally by various services and organizations, ranging from television stations to emergency services.

- **2. Role of the ITU-R:** The worldwide allocation of these frequencies is managed by the **International Telecommunications Unions - Radio (ITU-R)**.
- **3. Licensed Frequencies:** A licensed frequency is specifically allocated to a company or service, and only that specific entity is permitted to use it. Regulatory bodies are tasked with preventing interference between these different licensed bands.
- **4. License-Free Frequencies Definition:** These frequencies are available for use by anybody who wishes to use them, although their operation is still constrained by specific regulations.
- **5. License-Free Examples and Regulation:** Common license-free frequencies include 315MHz, 433MHz, 868MHz, 915MHz, and 2.4GHz. Regulations governing these bands restrict where they can be used and the duration of transmission to prevent interference.

Q11: How is the Extended Service Set (ESS) setup achieved in large organizations? Which existing wireless network topology does this setup utilize, and what common network uses this topology?

The Extended Service Set (ESS) is designed to provide wireless network coverage over a significant area, which is required for large organizations.

- **2. Setup Method:** The ESS setup is achieved by **strategically placing multiple Wireless Access Points (WAPs)** throughout the organization's space.
- **3. Coverage Mechanism:** Each individual WAP is configured to serve a specific coverage area, allowing the ESS to maintain continuous coverage.
- **4. Topology Utilized:** This kind of setup explicitly utilises the **Point-to-Multipoint (p2mp) Topology**.
- **5. Common Network Usage:** All **cellular networks** use this Point-to-Multipoint topology to ensure that consumers can access services without interruption even when they are moving.

Q12: List and briefly describe four major components of an ACL, such as those related to identifying the entry or defining network traffic rules.

An Access Control List (ACL) relies on several components to define its rules for allowing or denying access.

- **2. Sequence Number:** This component provides a specific numerical identifier for the individual ACL entry.
- **3. Statement:** This is the core rule definition, which dictates whether a source is to be **permitted** or **denied** access, often utilising a wildcard mask or address.
- **4. Network Protocol:** This component allows the administrator to permit or deny specific types of networking protocols, such as IP, TCP, UDP, or ICMP.
- **5. Source or Destination:** This component defines the IP address (or range of addresses) of either the destination of the traffic or the source, and can also be set to allow all addresses.

Q13: Explain how WPA3 defends against stronger brute force attacks compared to WPA2, particularly in the context of offline password guesses.

WPA3 was designed as an improved version of WPA2, featuring stronger safeguards against malicious guessing attacks.

- **2. Targeted Defense:** WPA3 specifically provides defense against the threat posed by **offline password guesses**.
- **3. Access Constraint:** It achieves this defense by limiting a user to only **one guess per attempt**.
- **4. Physical Requirement:** WPA3 requires the user (or attacker) to engage directly with the Wi-Fi equipment. This forces the attacker to be physically present and online each time they wish to attempt a password guess.
- **5. WPA2 Vulnerability:** In comparison, WPA2 often lacks built-in encryption and privacy features in public open networks, making offline brute force attacks a substantial security risk.

2-MARK QUESTIONS

Q14: What are the two types of media (waves) that a wireless network uses for data or voice communication?

- Wireless networks rely on waves to transmit data instead of wires.
- **Two Waves:** The two types of media used are **electromagnetic waves (EM Waves)** or **infrared waves**.

Q15: What specific type of wave communication is typically used when remote controls communicate with controlled devices in a Point-to-Point (p2p) topology?

- Remote controls are a classic example of p2p communication.
- **Wave Type:** Communication between remote controls and controlled devices typically takes place using **infrared (IR) waves**.

Q16: What is the primary disadvantage of using a Point-to-Point (p2p) topology related to data transmission mode?

- P2P networks define communication between two devices.
- **Disadvantage:** P2P networks are usually slow because they operate in **half duplex mode**, meaning data can only be sent by one device at a time.

Q17: In the context of 802.11 networks, what is the upstream wired network called?

- The 802.11 standard defines terminology for connections beyond the wireless clients.
- **Wired Network Term:** The 802.11 standard refers to the upstream wired network as the **distribution system (DS)**.

Q18: Name the WLAN standard term for the Multipoint-to-Point (mp-2-p) wireless setup.

- Mp-2-p is the common topology for small wireless networks.
- **WLAN Standard Name:** The wireless LAN standard name for the Multipoint-to-Point setup is the **Basic Service Set (BSS)**.

Q19: Which multiplexing technique is used by the Multipoint-to-Point (mp-2-p) topology to manage bandwidth effectively?

- Mp-2-p requires a method to manage bandwidth when multiple devices compete for airtime.
- **Technique:** The Mp-2-p topology uses **Frequency Division Multiplexing (FDM)** to manage bandwidth effectively.

Q20: What is the fundamental architecture unit of Bluetooth, and what is the typical transmission capacity of Bluetooth?

- Bluetooth is a short-distance communication standard.
- **Unit and Capacity:** The basic architecture unit of Bluetooth is a **piconet**, and its typical transmission capacity is **720 kbps**.

Q21: What is the purpose of the Service Discovery Protocol (SDP) Layer in the Bluetooth Protocol Stack?

- The Bluetooth stack includes several layers for various management tasks.
- **Purpose:** The Service Discovery Protocol (SDP) Layer allows the Bluetooth device to **discover the services available** on another Bluetooth-enabled device.

Q22: In the RTP Header Format, what is the function of the M (Marker) field?

- The RTP Header contains the single-bit 'M' field.
- **Function:** The 'M' (Marker) field is used by the application to **indicate the end marker** of its transmitted data.

Q23: In Enterprise Network Security, what is the networking architecture that allows for the classification and management of traffic using advanced ACL criteria?

- Advanced ACLs can control traffic based on criteria like Type of Service or priority.
- **Architecture:** The networking architecture that allows for the classification and management of traffic using advanced ACL criteria is the **Differentiated Services Code Point (DSCP)**.

Q24: What specific type of key does the WPA protocol use, which proved more difficult to crack than WEP's static key?

- **Deception (WPA Improvement):** WPA was designed to fix the vulnerabilities of WEP's static key encryption.
- **WPA Key Type:** WPA introduced the **Temporal Key Integrity Protocol (TKIP)**, which uses a dynamic 128-bit key that was more challenging to crack.

Q25: What is the minimum and maximum character length required for the plain-English password used to encrypt a network with WPA2-PSK?

1. WPA2-PSK requires a single, plain-English password for network access.
2. **Length Requirement:** The password used to encrypt a WPA2-PSK network must be between **8 and 63 characters long**.